



# Example 2: Mass Transfer Simulation in Recycled PP Bottles

## Synopsis

SFPPy Example: Simulating **1D mass transfer** of **toluene** from a **300  $\mu\text{m}$  recycled PP bottle** into **fatty liquid food**, with and without a **PET functional barrier (FB)**.

 **Example 2** showcases functional syntax (low-level), providing direct and transparent access to simulation details. If you prefer higher-level syntax with less coding effort and fewer dependencies, check out **Example 3** and **Example 4**, which leverage operator overloading and advanced methods for a more streamlined experience.

## Example 2: Mass Transfer Simulation in Recycled PP Bottles

### Overview

#### Simulation Steps

#### Expected Outcomes

### Dependencies

#### Installation

### Step 1: Define Contact Conditions

### Step 2: Define Bottle Geometry

### Step 3: Define Migrant (Toluene)

### Step 4: Define PP Bottle Walls

### Step 5: Define Liquid Food Properties

### Step 6: Run Migration Simulation (No Functional Barrier)

### Step 7: Add a PET Functional Barrier (FB)

### Step 8: Run Simulation with Functional Barrier

### Step 9: Compare Migration (With vs Without FB)

### Step 10: Study FB Thickness (2 $\mu\text{m}$ to 60 $\mu\text{m}$ )

### Step 11: Save and Print Figures

### Conclusion

## Overview

This simulation evaluates the **migration kinetics** of **toluene** from **recycled polypropylene (PP)** into a **fatty liquid food**. We also analyze the **effectiveness of a PET functional barrier (FB)** in reducing migration.



## Simulation Steps

#

- 1 Define bottle geometry: A 1L bottle with a **body** and **neck**.
- 2 Set up polymer layers: PP containing **toluene**, with/without PET FB.
- 3 Define liquid food properties: Fatty liquid, storage conditions.
- 4 Run mass transfer simulations:
  - ● Without a functional barrier.
  - ● With a 30  $\mu\text{m}$  PET FB.
- 5 Compare migration kinetics between both cases.
- 6 Perform a systematic study on **FB thickness** (2  $\mu\text{m}$  to 60  $\mu\text{m}$ ).
- 7 Save and print all simulation results.



## Expected Outcomes

#

- ✓ Analysis of migration kinetics (with and without FB).
- ✓ Determine optimal PET thickness for reducing migration.
- ✓ Generate plots and reports for food safety analysis.



## Dependencies

This example requires the following **SFPPy** modules:

|  Module |  Description |
|--------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| <code>patankar.loadpubchem</code>                                                          | Retrieves <b>molecular properties</b> from PubChem.                                             |
| <code>patankar.geometry</code>                                                             | Defines <b>packaging geometry</b> (e.g., bottles, cans).                                        |
| <code>patankar.food</code>                                                                 | Provides <b>food layer properties</b> for migration simulations.                                |
| <code>patankar.layer</code>                                                                | Defines <b>polymer materials</b> (PP, PET, etc.).                                               |
| <code>patankar.migration</code>                                                            | Implements <b>1D finite-volume solvers</b> for mass transfer.                                   |
| <code>matplotlib.pyplot</code>                                                             | Used for <b>plotting results</b> .                                                              |



## Installation

#

If not installed, ensure you have the SFPPy framework in your environment:

```
pip install sfppy
```



## Step 1: Define Contact Conditions

- Temperature: 20°C

- **Contact duration: 450 days**
- **Maximum initial concentration in polymer: 10 mg/kg**

```
contactTemperature = (20, "degC")
contactTime = (450, "days")
maxConcentration = 10 # mg/kg
```



## Step 2: Define Bottle Geometry

We model a **1L bottle** using the `Packaging3D` class.

```
from patankar.geometry import Packaging3D

bottle = Packaging3D(
    "bottle",
    body_radius=(40, "mm"),
    body_height=(0.2, "m"),
    neck_radius=(1.8, "cm"),
    neck_height=0.05
)
```

♦ **Computed values:**



**Volume = 1L**



**Contact surface area** = Calculated automatically



## Step 3: Define Migrant (Toluene)

Retrieve the **chemical properties** of **toluene** from **PubChem**:

```
from patankar.loadpubchem import migrant
surrogate = migrant("toluene")
```



## Step 4: Define PP Bottle Walls

Create a **300 µm PP layer** containing **toluene**:

```
from patankar.layer import PP

PPwalls_with_toluene = PP(
    l=(300, "um"),
    substance=surrogate,
    C0=maxConcentration,
    T=contactTemperature
)
```



## Step 5: Define Liquid Food Properties

Create a **fatty liquid food** model using inheritance:

```
import patankar.food as food

class liquidFood(food.realfood, food.liquid, food.fat):
    name = "liquidFood"

FOODlayer = liquidFood(
    volume=bottle.get_volume_and_area()[0],
    surfacearea=bottle.get_volume_and_area()[1],
    contacttime=contactTime,
    contacttemperature=contactTemperature
)
```



## Step 6: Run Migration Simulation (No Functional Barrier)

Run the **mass transfer simulation** for **toluene migration** from **PP** into **food**:

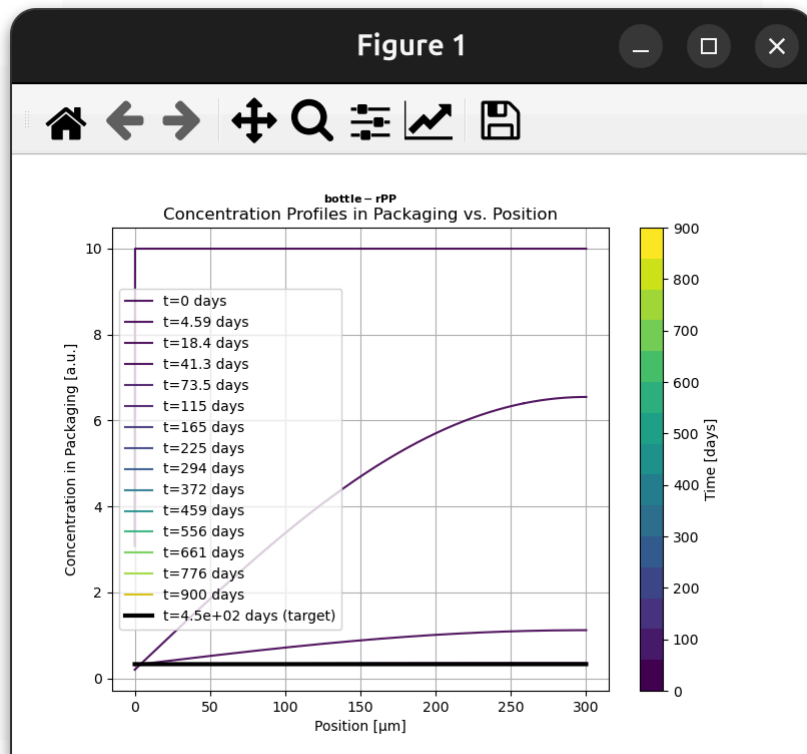
```
from patankar.migration import senspatankar as solver

ref_simulation = solver(
    PPwalls_with_toluene, # PP containing toluene
    FOODlayer,           # Fatty liquid food
    name="bottle-rPP"
)
```



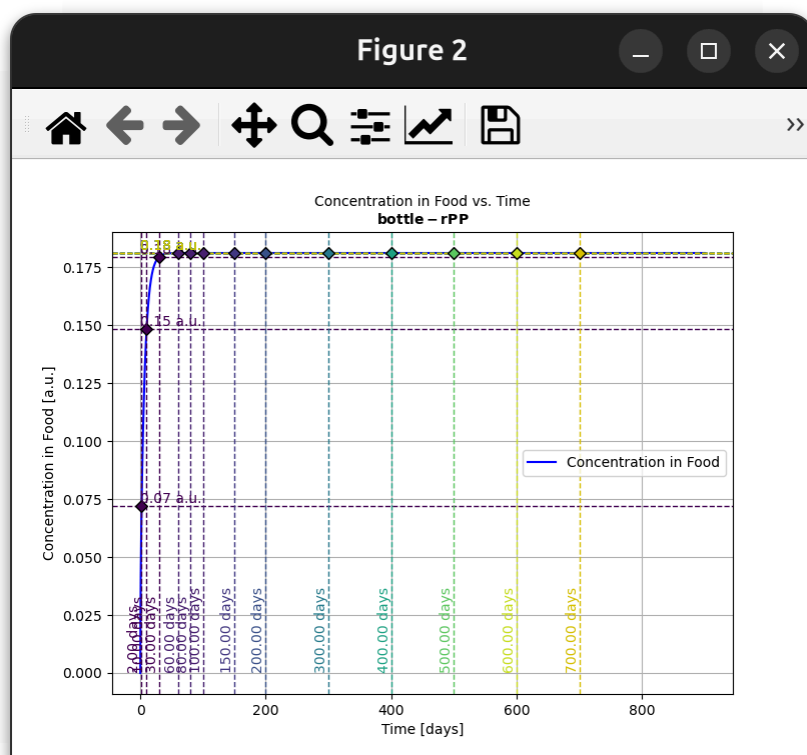
**Plot concentration profiles:**

```
ref_simulation.plotCx()
```



Plot migration kinetics:

```
ref_simulation.plotCF()
```





## Step 7: Add a PET Functional Barrier (FB)

Introduce a 30  $\mu\text{m}$  PET barrier layer:

```
from patankar.layer import gPET

PET_functionalBarrier = gPET(
    l=(30, "um"),
    substance=surrogate,
    C0=0, # Virgin PET (no migrant)
    T=contactTemperature
)

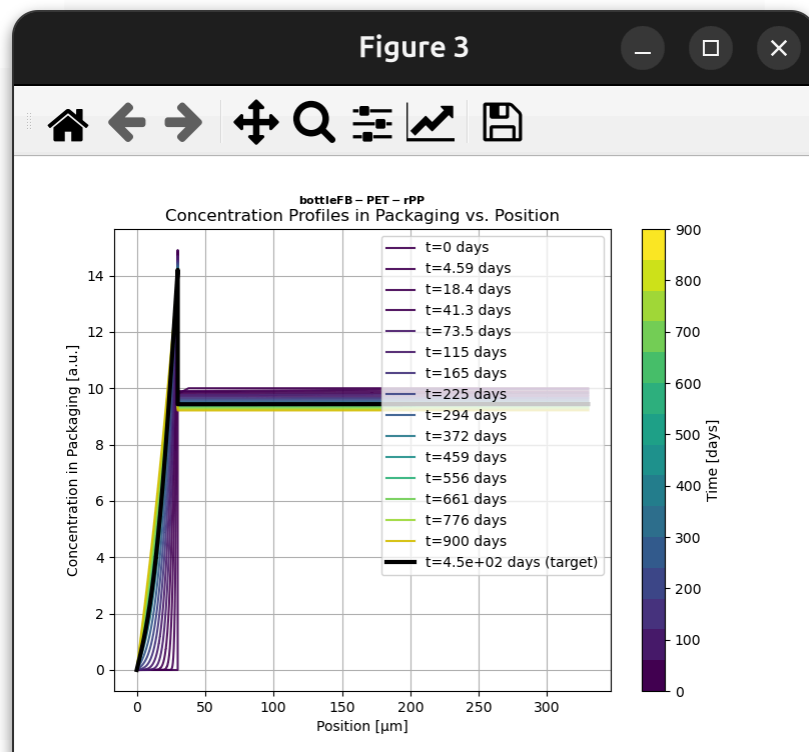
# Combine PET + PP
FBwalls_with_toluene = PET_functionalBarrier + PPwalls_with_toluene
```



## Step 8: Run Simulation with Functional Barrier

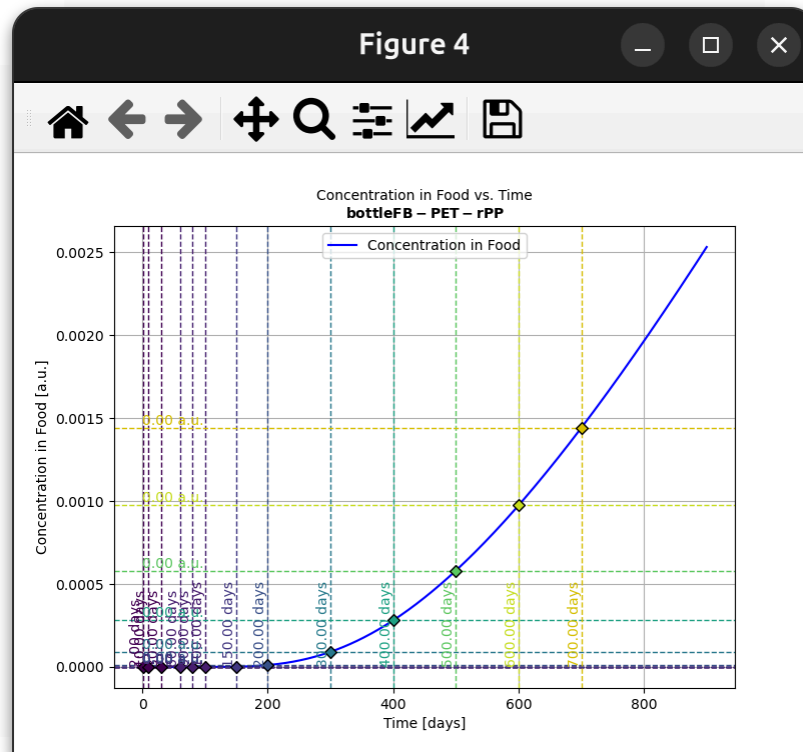
Re-run the simulation with the PET FB:

```
fb_simulation = solver(
    FBwalls_with_toluene,
    FOODlayer,
    name="bottleFB-PET-rPP"
)
fb_simulation.plotCx()
```



## Compare migration kinetics:

```
fb_simulation.plotCF()
```

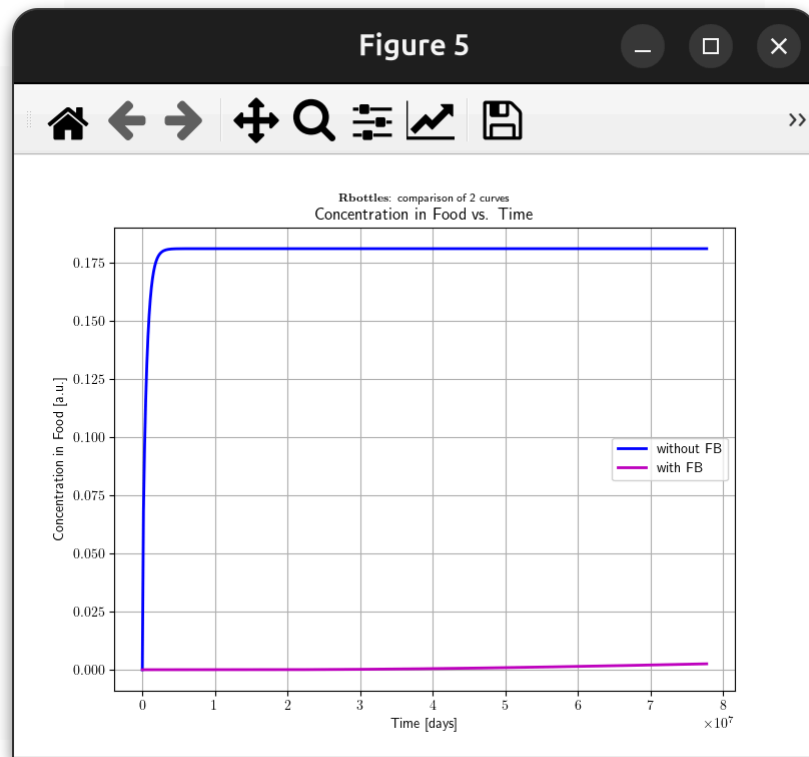


## Step 9: Compare Migration (With vs Without FB)

```
from patankar.migration import CFSimulationContainer as store

allCF = store(name="Rbottles")
allCF.add(ref_simulation, "without FB", "b") # Reference case (blue)
allCF.add(fb_simulation, "with FB", "m")     # Functional barrier case (magenta)

# Plot both cases
allCF.plotCF()
```



## Step 10: Study FB Thickness (2 $\mu\text{m}$ to 60 $\mu\text{m}$ )

```
fullcomparison = store(name="fb study")
fullcomparison.add(ref_simulation, "without FB", "b")

# Vary PET thickness from 2  $\mu\text{m}$  to 60  $\mu\text{m}$ 
for fb_thickness in range(2, 61, 4):
    print(f"Solving for FB = {fb_thickness}  $\mu\text{m}$ ")

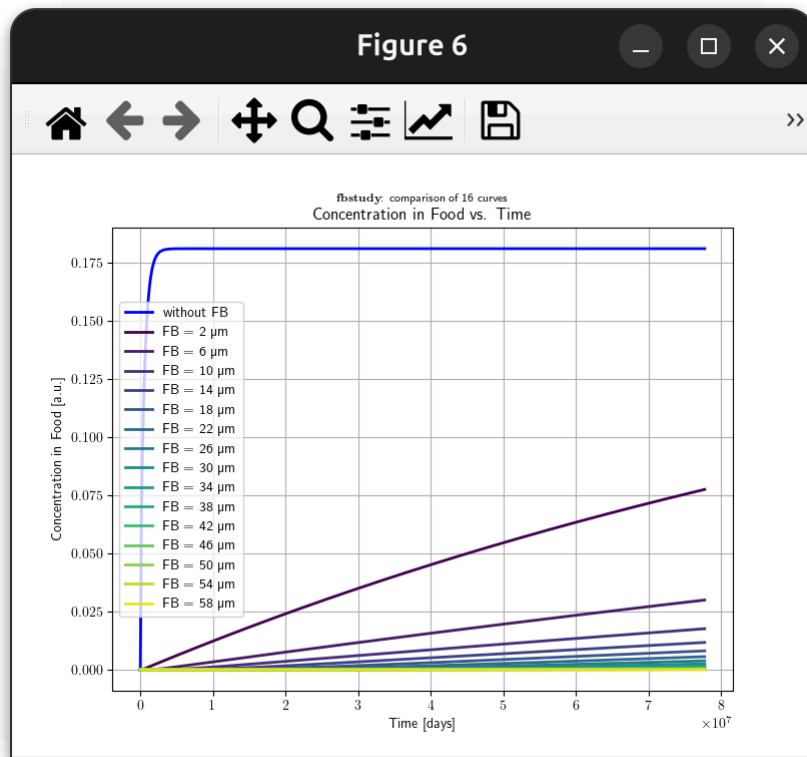
    FBwalls_with_toluene.l[0] = (fb_thickness, "um") # Update thickness

    sim = solver(FBwalls_with_toluene, FOODlayer, name=f"FB-{fb_thickness}um")

    fullcomparison.add(sim, f"FB = {fb_thickness}  $\mu\text{m}$ ")

#  Plot all FB thickness scenarios
fullcomparison.plotCF()
```





## Step 11: Save and Print Figures

```
from patankar.migration import print_figure

outputfolder = "tmp"
printconfig = {"destinationfolder": outputfolder, "overwrite": True}

print_figure(allCF.plotCF(), **printconfig)
print_figure(fullcomparison.plotCF(), **printconfig)
```

## Conclusion

- ✓ PET functional barriers reduce migration.
- ✓ Increasing PET thickness further reduces migration.
- ✓ SFPPy enables precise simulation and analysis.

 **Next Steps:** Extend the study with different **storage conditions**, **food types**, or **migrants**.

This document is based on **Example 2** from the **SFPPy** project. For more details, check the full documentation. 



# SFPPy for Food Contact Compliance and Risk Assessment

Contact [Olivier Vitrac](#) for questions | [Website](#) | [Documentation](#)